Complex Number

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1 Complex Number

I Complex number (複數)

i Complex Number

The set of complex numbers $\mathbb{C}$ is defined as

$$\mathbb{C} \coloneqq \{a + bi \mid a, b \in \mathbb{R}\},$$

where i is defined as the only complex number such that $i^2 = -1$ and is called the imaginary unit.

For a complex number $z = a + bi$ with $a, b \in \mathbb{R}$, a is called its real part (實部) and denoted as $\Re(z)$, $\mathrm{Re}(z)$, or $\mathcal{Re}(z)$, b is called its imaginary part (虛部) and denoted as $\Im(z)$, $\mathrm{Im}(z)$, or $\mathcal{Im}(z)$, z is called an imaginary number (虛數) if $b \neq 0$, and z is called a pure imaginary number (純虛數) if $a = 0$ and $b \neq 0$.

ii Addition, subtraction, and multiplication

For two complex numbers $y = a + bi$ and $z = c + di$ with $a, b, c, d \in \mathbb{R}$:

$$y + z = (a + c) + (b + d)i.$$

$$y - z = (a - c) + (b - d)i.$$

$$y \cdot z = (ac - bd) + (ad + bc)i.$$

iii Absolute value (絕對值), magnitude, radial distance (向徑), or modulus (模/模長)

The absolute value, magnitude, radial distance, or modulus of a complex number $z = a + bi$ with $a, b \in \mathbb{R}$ is $|z| = \sqrt{a^2 + b^2}$.

iv Complex conjugate (共軛)

The (complex) conjugate of a complex number $z = a + bi$ with $a, b \in \mathbb{R}$, denoted as $\bar{z}$ or z^* and called z bar, is defined as $a - bi$.

v Division

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}, z \neq 0$$

vi Properties

For complex numbers z, z_1, z_2:

$$|z| = \sqrt{z \cdot \bar{z}}$$

$$\overline{z_1} + \overline{z_2} = \overline{z_1 + z_2}.$$

$$\overline{z_1} - \overline{z_2} = \overline{z_1 - z_2}.$$

$$\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}.$$

$$\Re(\overline{z_1} \cdot z_2) = \Re(z_1 \cdot \overline{z_2}).$$

$$\frac{\overline{z_1}}{\overline{z_2}} = \overline{\left(\frac{z_1}{z_2}\right)}, \quad z_2 \neq 0.$$

$$\overline{z^n} = \left(\bar{z}\right)^n, \quad zn \neq 0 \wedge n \in \mathbb{Z}.$$

II Polynomials

i Fundamental theorem of algebra (代 數 基 本 定 理), d'Alembert's theorem, or the d'Alembert–Gauss theorem

Every non-constant single-variable polynomial with complex coefficients has at least one complex root.

Every non-zero, single-variable, degree n polynomial with complex coefficients has, counted with multiplicity, exactly n complex roots.

ii Equality of polynomials with real coefficients of complex number and its conjugate

For all polynomials with real coefficients f, for all complex numbers z:

$$\overline{f(z)} = f\left(\bar{z}\right).$$

Pair of imaginary roots (虛根成對定理): If a polynomial with real coefficients has an imaginary root, then its conjugate is also a root. Thus, an odd-degree polynomial with real coefficients must have at least one real root.

III Argument

i Argument (輻角)

For a complex number $z \neq 0$, the argument of z, denoted as $\arg(z)$, is definited as any real number φ such that

$$z = |z|e^{i\varphi}.$$

ii Principal argument (輻角主值/主輻角)

For a complex number $z =\neq 0$, the principal argument of z, denoted as $\operatorname{Arg}(z)$, is definited as the argument of z with in $(-\pi, \pi]$. (Some sources define it as the argument of z with in $[0, 2\pi)$.)

iii Polar form (極式)

The polar form of a complex number z is

$$z = |z|(\cos\theta + i\sin\theta), \quad \theta \in \mathbb{R}.$$

iv Properties

For any real number θ, φ, a, b:

$$\overline{e^{i\theta}} = e^{-i\theta}.$$

$$(ae^{i\theta})(be^{i\varphi}) = (ab)e^{i(\theta+\varphi)}.$$

$$\frac{ae^{i\theta}}{be^{i\varphi}} = \frac{a}{b}e^{i(\theta-\varphi)}.$$

IV Complex plane (複數平面), Argand plane (阿爾岡平面), or Gaussian plane (高斯平面)

i Complex plane, Argand plane, or Gaussian plane

The complex plane is the plane formed by the complex numbers, with a Cartesian coordinate system such that the horizontal x-axis, called the real axis, is formed by the real numbers, and the vertical y-axis, called the imaginary axis, is formed by the imaginary numbers.

ii De Moivre's formula (棣美弗公式)

$$(r(\cos\theta + i\sin\theta))^n = r^n(\cos(n\theta) + i\sin(n\theta)), \quad r \neq 0 \land n \in \mathbb{Z}.$$

iii Hyperbolic De Moivre's formula

$$(r(\cosh\theta + i\sinh\theta))^n = r^n(\cosh(n\theta) + i\sinh(n\theta)), \quad r \neq 0 \land n \in \mathbb{Z}.$$

iv Dot product

The dot product of two vectors in $\mathbb{R}^2$ corresponding to the two complex numbers z_1 and z_2 on the complex plane is $\Re(z_1 \cdot \overline{z_2})$.

V Geometric meaning of exponentiations of complex numbers on complex plane

i Raciporial of positive number exponentiation of 1

For $n \in \mathbb{N}$, $1^{\frac{1}{n}}$, that are, the n roots of $x^n = 1$, that are, $\omega = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ to the power of 0 to $(n-1)$, that are,

$$1^{\frac{1}{n}} = e^{i\frac{2\pi k}{n}}, \quad k \in \mathbb{N}_0 \land k < n,$$

are the n vertices of a regular n-gon inscribed in a unit circle centered at the origin including 1 on the complex plane.

ii Raciporial of positive number exponentiation of nonzero complex number

For $n \in \mathbb{N}$ and $z \in \mathbb{C}_{\neq 0}$, $z^{\frac{1}{n}}$, that are, the n roots of $x^n = z$, that are, $\sqrt[n]{|z|}$ multiplied by $\omega = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ to the power of 0 to $(n-1)$, that are,

$$z^{\frac{1}{n}} = \sqrt[n]{|z|}e^{i\frac{\mathrm{Arg}(z)+2\pi k}{n}}, \quad k \in \mathbb{N}_0 \land k < n,$$

are the n vertices of a regular n-gon inscribed in a circle of radius $\sqrt[n]{|z|}$ centered at the origin including $\sqrt[n]{|z|}e^{i\frac{\operatorname{Arg}(z)}{n}}$ on the complex plane.

iii Nonzero complex number exponentiation of nonzero complex number

For $w, z \in \mathbb{C}_{\neq 0}$, z^w, that are,

$$\begin{aligned}
z^w &= e^{w \ln(z)} \\
&= \left(|z|e^{i\left(\operatorname{Arg}(z)+2\pi k\right)}\right)^w \\
&= |z|^{\Re(w)}|z|^{\Im(w)i}e^{i\Re(w)\left(\operatorname{Arg}(z)+2\pi k\right)}e^{-\Im(w)\left(\operatorname{Arg}(z)+2\pi k\right)} \\
&= |z|^{\Re(w)}e^{-\Im(w)\left(\operatorname{Arg}(z)+2\pi k\right)+i\left(\Re(w)\left(\operatorname{Arg}(z)+2\pi k\right)+\Im(w)\ln|z|\right)}, \\
&\quad k \in \mathbb{Z},
\end{aligned}$$

are points with equal angle difference $\Re(w)2\pi$ on the spiral

$$|z|^{\Re(w)}e^{-\Im(w)\theta}\left(\cos\left(\Re(w)\theta + \Im(w)\ln|z|\right) + i\sin\left(\Re(w)\theta + \Im(w)\ln|z|\right)\right)$$

of parameter $\theta \in \mathbb{R}$ including that of $\theta = \operatorname{Arg}(z)$ on the complex plane.

Modulus:

$$\left|z^w\right| = |z|^{\Re(w)}e^{-\Im(w)\left(\operatorname{Arg}(z)+2\pi k\right)}.$$

- When $\Im(w) > 0$: The modulus decays exponentially with respect to k, approaching zero as k increases and diverging to infinity as k decreases.
- When $\Im(w) < 0$: The modulus increases exponentially with respect to k, diverging to infinity as k increases and approaching zero as k decreases.
- When $\Im(w) = 0$: The modulus is always $|z|^{\Re(w)}$, lying on a circle with radius $|z|^{\Re(w)}$ centered at the origin.

Angle (parameter θ):

$$\theta(k) = \Re(w)\left(\operatorname{Arg}(z) + 2\pi k\right) + \Im(w)\ln|z|.$$

- When $\Re(w) > 0$: The angle increases linearly with respect to k, rotating counterclockwise as k increases and clockwise as k decreases.
- .When $\Re(w) < 0$: The angle decreases linearly with respect to k, rotating clockwise as k increases and counterclockwise as k decreases.
- When $\Re(w) \in \mathbb{Z}$: The angle is always $\left(\Re(w)\operatorname{Arg}(z) + \Im(w)\ln|z|\right) \mod (2\pi)$, lying one the ray with angle $\Re(w)\operatorname{Arg}(z) + \Im(w)\ln|z|$.

Spiral direction:

- When $\Re(w)\Im(w) > 0$: The root lies in a clockwise spiral centered at the origin on the complex plane.
- When $\Re(w)\Im(w) < 0$: The root lies in a counterclockwise spiral centered at the origin on the complex plane.

- When $\Im(w) = 0$: The root lies in a circle centered at the origin on the complex plane.

Number of distinct roots:

- When $\Im(w) = 0 \wedge \Re(w) \in \mathbb{Q}$: Suppose $|\Re(w)| = \frac{m}{n}$ with $n \in \mathbb{N}$ and $\gcd(m,n) = 1$, there are n distinct roots, which are the n vertices of a regular n-gon inscribed in a circle of radius $|z|^{\Re(w)}$ centered at the origin including $|z|^{\Re(w)}e^{i\Re(w)\,\mathrm{Arg}(z)}$ on the complex plane.
- Otherwise: There are countably infinitely many distinct roots.

VI Extended complex number

$$\{\bar{\mathbb{C}} = \{a + bi \mid a, b \in \bar{\mathbb{R}}\}.$$